Battery: choosing between samples

Inputs: formulation context and the first 50 charge/discharge cycles.

Task: forecast retention at cycles 351–360; select the sample with higher retention in each pair.

Bars: mean of completed runs. Whiskers: observed min–max, not confidence intervals.

Opus: 2 Anthropic-route + 1 Google-route run. † Format adaptation. Unequal run counts.

Trend baseline extends the recent capacity trend. 10 traces, 12 related comparisons.

Catalyst: predicting current

Inputs: current–voltage scans ending at 0.600 V.

Task: predict signed current at 0.800 V and choose the higher value in each pair.

Bars: mean of completed runs. Whiskers: observed min–max, not confidence intervals.

Opus: 2 Anthropic-route runs; Google had no scored answer. † Format adaptation.

Linear-tail baseline extends the scan's final segment. 8 scans, 7 related comparisons.